

Human Analyst vs. AI Analyst

Comparing ChatGPT's Data Analyst with my own Excel analysis of 6,607 student records

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Approach: same data, two analysts

Kaggle Student Performance Factors dataset: 6,607 students, 20 variables (behavior, home, school, and outcome data).

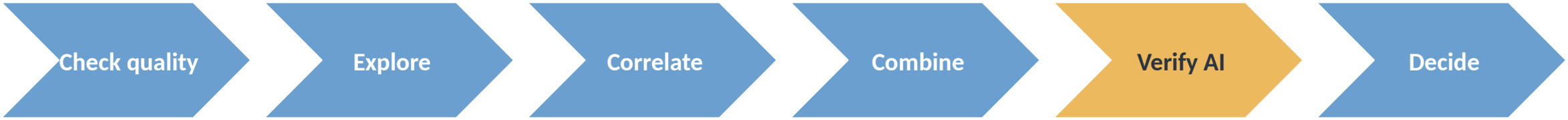
My analysis (Excel)

- Quality checks: COUNTBLANK, MIN/MAX
- CORREL, RSQ, LINEST, AVERAGEIFS
- Pivot table + 4 charts (data fusion)

ChatGPT Data Analyst

- Uploaded the original Kaggle file
- Course prompt + 3 follow-up prompts
- Python cleaning, charts, regression models

A systematic process to reduce uncertainty



Each step removes a source of error before the next one, so the final decision rests on checked evidence instead of a single tool's output.

Data quality: both found the same problems

We agreed on what was wrong with the data, but made different choices about how to fix it.

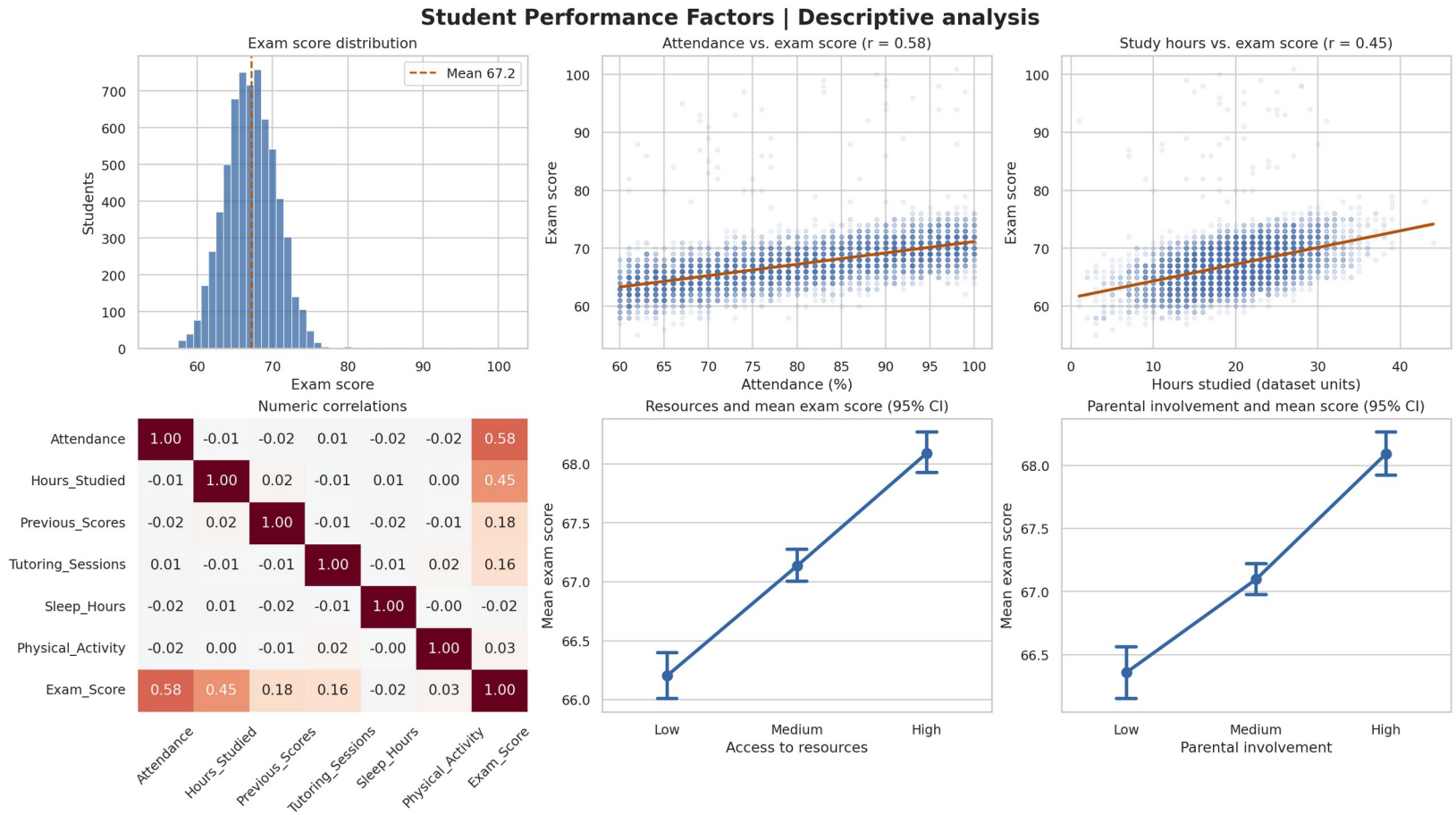
Quality check	My Excel analysis	ChatGPT Data Analyst
Duplicate records	0 found	0 found
Missing values	235 in 3 columns; left blank	235 values in 229 records; labeled "Unknown"
Exam score of 101	Removed (impossible on a 100-point scale)	Kept and flagged (could be extra credit)
Outliers	Not checked	104 by IQR rule (76 high, 28 low)



Why it matters: one bad value can change results and decisions. GPT's "keep, flag, and test both ways" approach is more transparent; my approach was simpler but should have been checked with the grading policy.

What ChatGPT found: the statistical story

GPT produced a six-panel analysis in about one minute.



r = 0.58
Attendance is the strongest link to exam score

r = 0.45
Study hours come second

+1.9 pts
High vs. low access to resources

≈ 0
Sleep and physical activity barely related

Source: ChatGPT Data Analyst output (OpenAI, 2026), student_performance_visualizations.png.

Side by side: the numbers matched

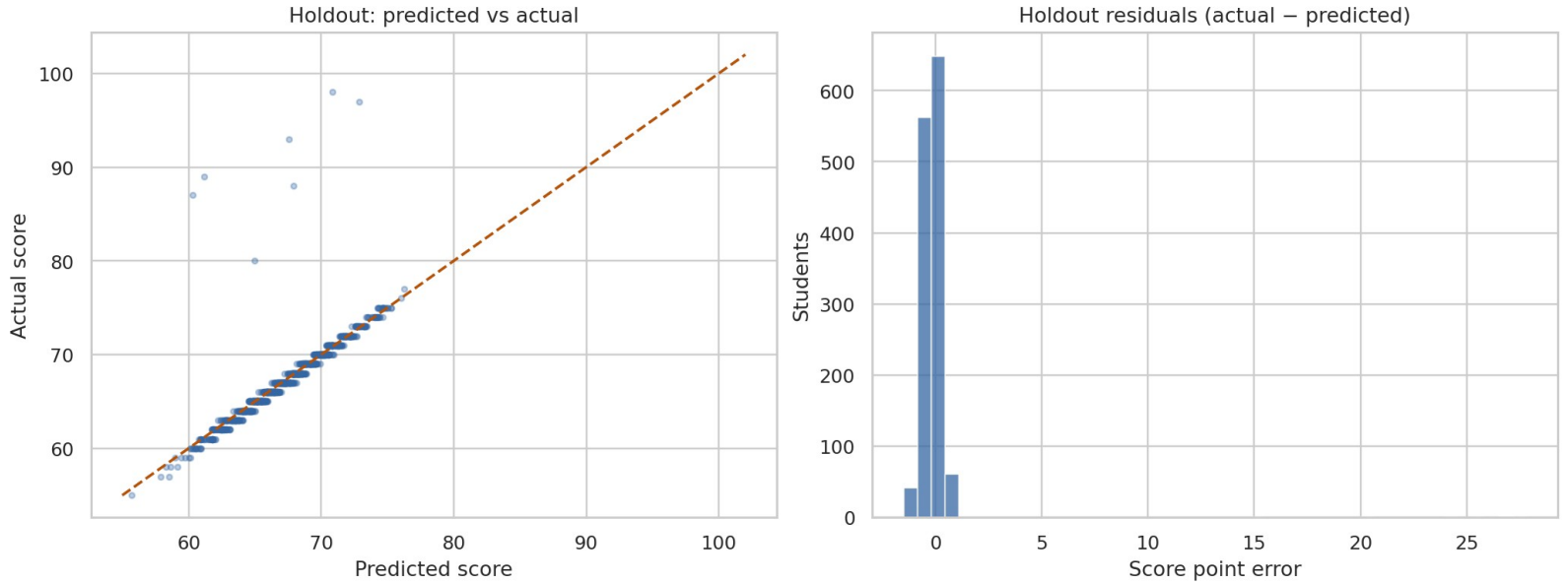
Tiny differences come from one choice: I removed the score of 101; GPT kept it.

Result	My Excel analysis	ChatGPT	Match?
Attendance correlation (r)	0.582	0.581	✔ Yes
Study hours correlation (r)	0.447	0.445	✔ Yes
Attendance + study hours together (R^2)	54.4%	54.1%	✔ Yes
Avg score, 60–69% vs 90–100% attendance	64.2 → 70.1	64.2 → 70.1	✔ Yes
High vs. low resources	68.1 vs 66.2	68.1 vs 66.2	✔ Yes
Attendance × parental involvement	+~2 pts at every attendance level	Not combined unless asked	Only me
Predictive model	Not built	Ridge regression, R^2 0.77	Only GPT

I verified GPT's math: its regression (45.6 + 0.197 × Attendance + 0.293 × Hours) and its 68.2 prediction for 85% attendance and 20 hours both checked out.

Where GPT went further: a predictive model

Ridge regression on an 80/20 train/test split. Strong on paper, but with warning signs.



0.77 R^2 on test data

0.46 pts average error (MAE)

27.8 pts largest single error

Source: ChatGPT Data Analyst output (OpenAI, 2026), student_model_diagnostics.png.

My critique: scores sit almost perfectly on the line, which is unusual for real students and suggests a generated (synthetic) dataset. The result comes from one split with no cross-validation, and a few students are badly mispredicted. GPT flagged the last two points itself, but it never questioned whether the data was realistic.

What worked well, and where errors appeared

The AI was fast and accurate on the math, but it still needed a human to check its reasoning.

Worked well

- Full analysis in about 1 minute
- Every key number I checked was correct
- Transparent cleaning: flagged instead of deleting
- Warned that correlation is not causation
- Suggested fairness checks by gender, income, and disability
- Gave useful follow-up prompts

Errors and gaps

- Contradicted itself: said attendance and study hours "may be related," then found r = −0.01
- Called r = 0.58 "moderate" while the course treats ≥0.5 as strong
- Left the 101 score undecided
- Did not question whether the data is realistic
- Combined data types only when asked

Ethics, faith, and the business decision

Using AI responsibly turns fast analysis into decisions people can trust.

Privacy

This dataset is public and anonymous. Real student records should not be uploaded to a public AI tool.

Fairness

Models must be checked across gender, income, and disability groups so no group is treated unfairly.

Test everything

"But test them all; hold on to what is good" (1 Thessalonians 5:21). Verify AI output before acting on it.

Business solution for a school

- Flag students below 80% attendance for early outreach (scores rise ~6 points from the lowest to highest attendance band)
- Encourage 20+ study hours a week and involve parents
- Pilot first and compare with a similar group, because correlation alone does not prove cause

How to improve next time, and conclusions

The best result comes from combining AI speed with human judgment.

Improvements

1. Give the AI the grading rules up front (for example, scores max out at 100)

2. Ask for 5-fold cross-validation and a sensitivity test (101 kept vs. removed)

3. Ask it to combine numeric and categorical data (data fusion) from the start

4. Re-check every key number in Excel before using it

Conclusions

- AI is an excellent first-draft analyst: fast, thorough, and accurate on the math
- Humans are still needed for judgment: cleaning choices, realism checks, and causation
- High-quality data matters more than the tool used
- Verify, then decide

References

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Use of AI: ChatGPT Data Analyst performed the AI analysis compared in this presentation. Claude (Anthropic) helped organize the requirements, check ChatGPT's numbers against the data, and draft the slides, which I reviewed and edited.